SATYAJIT S

B.Tech. Computer Science • IIITDM Kancheepuram
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EDUCATION

Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram (IIITDM) *2024 – 2028*
Bachelor of Technology in Computer Science *CGPA: 8.90*

The Lords International School, Chennai *2023 – 2024*
Cambridge Board (A-Levels) *Percentage: 80.8%*

TECHNICAL SKILLS

- **Robotics:** ROS2, MoveIt2, Gazebo, RViz2
- **Machine Learning & Computer Vision:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, LangChain, LangGraph
- **Programming Languages:** C, C++, Python, JavaScript
- **Web Technologies:** HTML, CSS, React.js, Node.js, Express.js, REST APIs, PostgreSQL, Supabase, Pinecone
- **Tools:** Git, Linux, Docker

EXPERIENCE

e-Yantra Robotics Research Intern | IIT Bombay *May 2026 – Jul 2026*
- Deployed Vision-Language-Action (VLA) models on SO-101 bimanual arms to execute complex manipulation tasks, including precision pick-and-place and cube stacking.
- Collected expert demonstration datasets and trained imitation learning policies utilizing ACT (Action Chunking with Transformers) and smolVLA.

MaRS Student Rover Society (IIITDM) | Software Team Member *Jan 2025 – Present*
- Developed a Cartesian control pipeline for a 5-DOF manipulator using MoveIt2 to enable dynamic kinematics and precise end-effector control.
- The team secured 10th position at the International Rover Challenge (IRC).

PROJECTS UNDERTAKEN

Autonomous Holonomic Bot | e-Yantra Robotics Competition (IIT Bombay) *Sep 2025 – May 2026*
- Developed an autonomous holonomic robot using PID-controlled omnidirectional drive for precise trajectory tracking and docking.
- Integrated an ArUco marker vision pipeline for real-time pose estimation to execute automated object detection and pick-and-place tasks.
- Placed among the top 20 teams out of 300+ participating teams across India.

TD3-based Autonav | Reinforcement Learning Navigation *Sep 2026 – Oct 2026*
- Developed a TD3 reinforcement learning agent for TurtleBot3 maze navigation in Gazebo (ROS2 Humble, Stable-Baselines3) using LiDAR and relative-goal observations, with BFS geodesic reward shaping during training.
- Trained on a single maze and generalized to unseen maze layouts, achieving ~98% success on the training map.

Grayscale2RGB | Deep Learning Image Colourization *Jun 2026 – Jul 2026*
- Developed a conditional Denoising Diffusion Probabilistic Model (DDPM) using a custom UNet with spatial self-attention for facial image colourization.
- Applied Classifier-Free Guidance (CFG) during inference to control colour saturation and improve output quality.

CERTIFICATIONS

- Self-Driving – Map & Localization (Udemy) | Self-Driving – Odometry and Control (Udemy) | Python for Data Science and Machine Learning (Udemy)